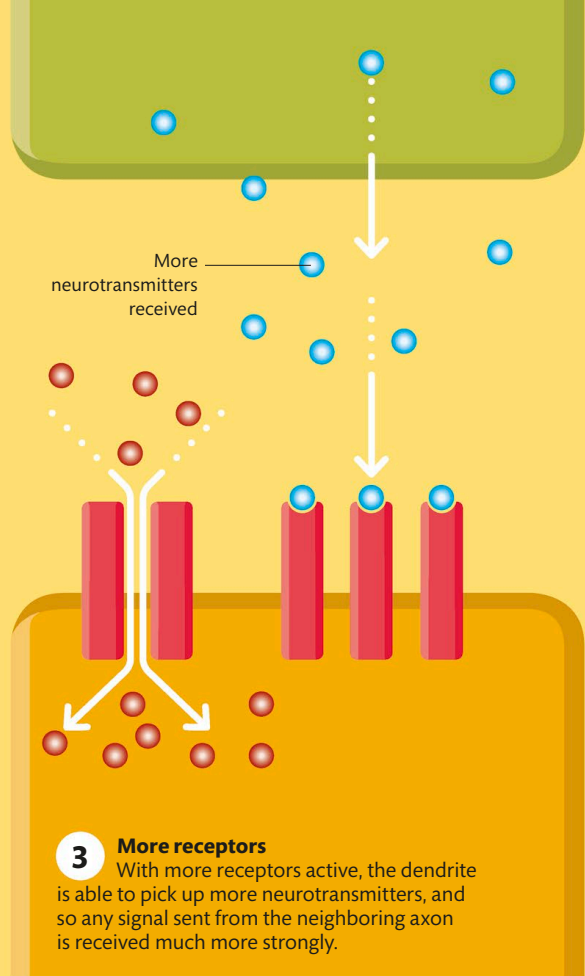
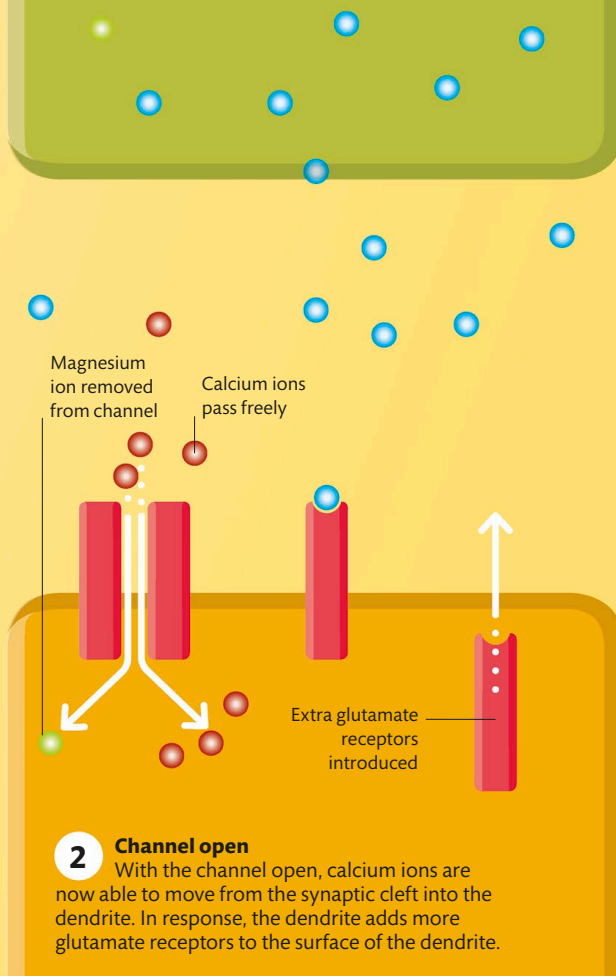
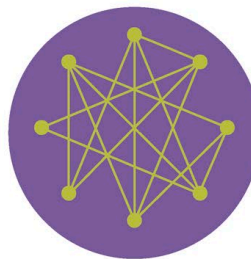




Small-world networks

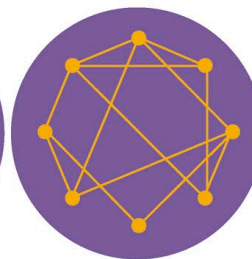
Brain cells are not connected in a regular pattern, nor are they in a random network. Instead, many of them exhibit a form of small-world network, where cells are seldom connected to their immediate neighbors but to nearby ones. This way of networking allows each cell to, on average, connect to any other in the smallest number of steps.

IT IS ESTIMATED THAT
THE HUMAN BRAIN
CONTAINS 100 TRILLION
CONNECTIONS BETWEEN
ITS 86 BILLION NEURONS



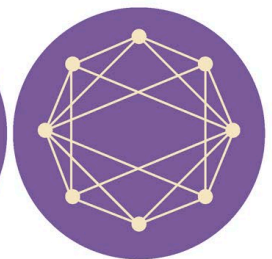
Random

A random network is good at making long-distance connections but poor at linking nearby cells.



Small-world

Small-world networks have good local and distance connections. Every cell is more closely linked than in the other two systems.



Lattice

By connecting every cell to its neighbors, this network has reduced scope to make long-distance connections.